

# YIKE ZHANG

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## RESEARCH INTERESTS

Reinforcement learning and sequential decision-making; human-in-the-loop optimisation; simulation-based policy learning; AI for sustainable mobility.

## EDUCATION

**University College London (UCL) — MSc Artificial Intelligence for Sustainable Development** *Sept 2025 – Sept 2026*

- Relevant modules: Probabilistic Modelling; Deep Representations and Learning; Statistical Natural Language Processing; Affective Computing and Human–Robot Interaction; Applied AI.

**University of Southampton — BEng Electronic Engineering** *Sept 2022 – Jun 2025*

- Awarded with Second Class Honours (Upper Division); Foundations of Machine Learning: 83/100.
- Relevant modules: Foundations of Machine Learning; Advanced Programming; Guidance, Navigation & Control; Control & Communications; Digital Systems & Microprocessors.

## RESEARCH & PROJECT EXPERIENCE

**MSc Dissertation — Human-in-the-Loop Bayesian Optimisation of Eco-Driving Feedback** *2026*

*Supervisor: Dr Mark Colley, UCL Interaction Centre*

- Applied human-in-the-loop multi-objective Bayesian optimisation to personalise eco-driving HUD configurations for individual drivers, jointly optimising an energy proxy, task load (NASA-TLX) and acceptance (van der Laan) over a 7-parameter design space.
- Ran the full participant study: 16 drivers, 21 evaluation rounds each (336 observations), with recommended configurations drawn from each driver’s own Pareto set.
- Built the experimental platform in Unity: a VR driving environment combining Google Photorealistic 3D Tiles (via Cesium) with a drivable road network generated from OpenStreetMap (418 road segments, 28.7 km, 564 junctions).
- Implemented a road-load vehicle energy model (rolling/aerodynamic resistance, EV regeneration) for real-time eco-scores, and an end-to-end automated experiment loop (round control, questionnaires, optimiser bridge) with scripted R analysis.

**Undergraduate Dissertation — Wearable Device for Badminton Stroke Recognition** *Oct 2024 – Mar 2025*

- Built a wrist-worn device (Arduino Nano 33 BLE Sense Rev2, 9-axis IMU, heart-rate/SpO2 sensor, 3D-printed 30 g enclosure) streaming attitude data over BLE with complementary-filter fusion.
- Classified four stroke types with an SVM (77.37% accuracy, macro-F1 0.76); predicted labels fed to the GPT-4o API for coach-style feedback, served on a Flask web interface.

**Undergraduate engineering projects** *2023 – 2024*

- Smart meter for microgrid management in critical infrastructure (Atmega644p); mixed-signal IC design incl. ring oscillator, 9-bit sequence recogniser and asynchronous serial interface (team of 6).

## PATENTS

- “A Positioning Component for Sensor Manufacturing” — Chinese utility-model patent, CNIPA, accepted Oct 2024.
- “An Adjustable Optical Projection Device” — Chinese utility-model patent, CNIPA, granted Aug 2025.

## PROFESSIONAL EXPERIENCE

**Inspur Co., Ltd. — Intern, Network Technology Support (server & big-data platform maintenance, Wireshark packet analysis)** *Jul – Sept 2024*

**Inner Mongolia Tongli Intelligent Technology — Intern, R&D Department (water-sensor inspection project)** *Jul – Sept 2023*

**Hohhot No.2 High School — Teaching Assistant** *Sept 2023 & Sept 2024*

## TECHNICAL SKILLS

**Machine Learning:** Python, PyTorch, Bayesian optimisation, probabilistic modelling, NLP fundamentals

**Simulation:** Unity (C#), Cesium, OpenStreetMap tooling, MATLAB

**Systems & Embedded:** C/C++, SystemVerilog/HDL, Arduino, Raspberry Pi, Xilinx FPGA, PCB design

**Tools:** Git, CMake, Jupyter, LTspice, ModelSim, Quartus

## LANGUAGES

Mandarin Chinese (native); English (fluent, full English-medium instruction since 2022).